

EXPERIMENT NO : 06

Time Series Analysis of Website Traffic

PROGRAM:-

```
import pandas as pd

import matplotlib.pyplot as plt

data = pd.read_csv('website_traffic.csv', parse_dates=['Date'])

data.set_index('Date', inplace=True)

data['Page_Views_Z'] = (

    data['Page_Views'] - data['Page_Views'].mean()

) / data['Page_Views'].std()

anomalies = data[data['Page_Views_Z'].abs() > 3]

plt.figure(figsize=(12, 5))

plt.plot(

    data.index,

    data['Page_Views'],

    label='Page Views'

)

plt.scatter(

    anomalies.index,

    anomalies['Page_Views'],

    color='red',

    label='Anomalies'

)

plt.legend()

plt.xlabel('Date')

plt.ylabel('Page Views')

plt.show()
```

OUTPUT:-

